# MODULE 4

## Quantitative Risk Assessment Using the Indicators & Warnings (I&W) Technique CIBC's Acquisition of a Digital Insurance Platform

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## 1. Overview: From Qualitative Prioritisation to Quantitative Assessment

Effective risk management requires both qualitative and quantitative analysis. Qualitative assessment helps organisations identify and prioritise risks, while quantitative techniques provide a more rigorous understanding of their probability, potential impact, and financial significance. Following the Module 3 qualitative assessment, three risks were identified as the most significant within CIBC's proposed acquisition of a digital insurance platform: R1 Data Privacy Breach (Score = 72), R2 Regulatory and Compliance Exposure (Score = 56), and R3 First-Mover Market Capture (Score = 56, Positive Risk).

These risks were selected because they represent the most significant threats and opportunities affecting CIBC's competitive position. A data privacy breach could undermine customer trust and strengthen competitors such as Lemonade and PolicyMe. Regulatory delays could allow TD Bank and RBC to establish market leadership before CIBC completes implementation. Conversely, successful first-mover market capture could generate substantial revenue growth, increased market share, and long-term competitive advantage.

To quantitatively assess these risks, the Indicators and Warnings (I&W) framework was applied. The framework uses measurable indicators and warning thresholds to identify deteriorating conditions before risks materialise. Monte Carlo simulation, sensitivity analysis, and Expected Monetary Value (EMV) analysis were used to estimate risk probability and impact, while machine learning techniques were considered as potential enhancements for predictive monitoring.

### 1.1 Data Source Note

The quantitative models were developed using data and assumptions derived from the Group 7 Risk Register, Risk Calculation Sheet, RT&RP documentation, and supporting project datasets. These sources provided the inputs required for Monte Carlo Simulation, Sensitivity Analysis, and Expected Monetary Value (EMV) calculations.

## 2. Quantitative Risk Assessment Results

### 2.1 R1 — Data Privacy Breach

R1 was assessed using Monte Carlo simulation supported by the I&W framework. Key indicators included unpatched critical vulnerabilities, anomalous access alerts, integration security scan pass rates, and unresolved penetration test findings. Three cost scenarios were developed from the risk register and modelled using a triangular distribution across 10,000 simulation iterations. Probability and impact were estimated by defining optimistic, most likely, and pessimistic loss scenarios and simulating a range of possible financial outcomes.

The analysis produced a mean expected loss of CAD $82.8M, with a 90th percentile loss of CAD $118.7M and a 25.6% probability that losses would exceed CAD $100M. These results demonstrate that a successful breach during platform integration could create significant financial, operational, and reputational consequences.

Based on these findings, the RT&RP was updated to establish a contingency reserve of CAD $119M and a Board stress-test benchmark of CAD $128M. As a potential machine learning enhancement, an Isolation Forest anomaly detection model could be trained on historical SOC alert data to identify abnormal access patterns before a breach occurs and provide an additional early-warning capability.

### 2.2 R2 — Regulatory and Compliance Exposure

R2 was assessed using sensitivity analysis within the I&W framework. Indicators included unresolved compliance findings, provincial licensing progress, Bill C-27 readiness, and outstanding OSFI Guideline B-10 requirements. A base impact of CAD $56M was established, and five key impact drivers were varied by ±30% to determine their influence on the overall risk outcome**.** Probability and impact were estimated by measuring how changes in key regulatory variables affected the overall financial exposure associated with compliance failure.

The analysis identified the likelihood of a regulatory halt as the most influential variable, generating an impact swing of ±CAD $11.8M. Regulatory fine severity and provincial launch delays were the second and third most influential drivers. These findings indicate that compliance readiness is the most important determinant of successful market entry.

As a result, the RT&RP prioritises compliance-related KRIs, enhanced regulatory monitoring, and dedicated compliance oversight. A gradient-boosting classification model was identified as a potential enhancement for estimating regulatory halt probability using compliance findings, licensing delays, and regulatory inquiry data.

### 2.3 R3 — First-Mover Market Capture

R3 was assessed using Expected Monetary Value (EMV) analysis. Key indicators included digital insurance adoption rates, competitor launch activity, advisor licensing levels, and new policy application volumes. Four adoption scenarios were developed and assigned probabilities using evidence from the risk register and external market research**.** Probability and impact were estimated by assigning likelihoods to alternative market adoption outcomes and calculating their expected financial value using the EMV framework.

The analysis produced a total EMV of CAD $187.1M, comprising CAD $168.0M from the Strong First-Mover scenario, CAD $17.8M from Partial First-Mover, CAD $3.0M from Late Entry, and −CAD $1.8M from Failed Launch. The positive EMV demonstrates that the expected benefits of capturing first-mover advantage substantially outweigh the potential downside risks.

Consequently, the RT&RP prioritises active opportunity exploitation through advisor training, marketing investment, and accelerated licensing initiatives. A time-series forecasting model, such as LSTM, was identified as a potential enhancement for predicting future adoption rates and providing earlier warning of deteriorating market performance

***Table 1: Summary of Quantitative Risk Assessment Results and Updated RT&RP Actions. Source: Group 7 Risk Register, Risk Calculation Sheet, and RT&RP Documentation, ALY6130 Risk Management Analytics, Spring 2026.***

| Risk | Method | Key Quantitative Output | Updated Treatment / RT&RP Action | ML Enhancement (Hypothetical) |
|---|---|---|---|---|
| **R1**<br>Data Privacy Breach<br>Score = 72<br>**Priority: HIGH** | Monte Carlo Simulation<br>Triangular distribution<br>10,000 iterations | Mean expected loss: CAD $82.8M<br>50th percentile: CAD $81.0M<br>90th percentile: CAD $118.7M<br>95th percentile: CAD $127.8M<br>P(Loss > CAD $100M): 25.6% | **Contingency reserve: CAD $119M (90th pct)**<br>Board stress-test benchmark: CAD $128M (95th pct)<br>Deploy zero-trust architecture and 24/7 SOC pre-launch<br>Complete Privacy Impact Assessment before go-live | Isolation Forest anomaly detection model on SOC alert patterns as real-time leading I&W indicator |
| **R2**<br>Regulatory & Compliance Exposure<br>Score = 56<br>**Priority: HIGH** | Sensitivity Analysis<br>Tornado Chart<br>5 key drivers<br>±30% variation | Base impact: CAD $56M<br>Top driver: Likelihood of Regulatory Halt (±CAD $11.8M)<br>Second driver: Regulatory Fine Severity (±CAD $7.7M)<br>Third driver: Provincial Launch Delay Duration (±CAD $6.3M) | Direct monitoring resources to compliance findings KRI as highest-sensitivity variable<br>Appoint dedicated compliance officer<br>Deploy automated regulatory monitoring tools<br>Quarterly PIPEDA/Bill C-27 legal reviews | Gradient boosting classifier predicting halt probability from compliance finding counts and licence delays |
| **R3**<br>First-Mover Market Capture<br>Score = 56<br>**Priority: HIGH – Positive Risk** | Expected Monetary Value (EMV)<br>4 adoption scenarios<br>EMV = P × Net Value | **Total EMV: CAD $187.1M**<br>Strong First-Mover: CAD $168.0M<br>Partial First-Mover: CAD $17.8M<br>Late Entry: CAD $3.0M<br>Failed Launch: −CAD $1.8M<br>Amber breach → −CAD $150M value<br>Red breach → −CAD $165M value | **Actively pursue opportunity; CAD $187.1M EMV justifies continued strategic investment and resource allocation**<br>Amber warning triggers emergency marketing review<br>Red warning triggers CEO/Board strategy reset<br>Complete advisor provincial licensing within 6 months | LSTM time-series forecasting model generating monthly adoption rate predictions from CIBC digital banking engagement data |

## 3 Conclusion

The quantitative assessment confirms the significance of all three risks. R1 presents the greatest downside exposure, with losses potentially exceeding CAD $100M in a substantial proportion of simulation outcomes. R2 demonstrates that regulatory readiness is the most critical factor influencing implementation success and market entry timing. R3 represents a major strategic opportunity, with an estimated value of CAD $187.1M if CIBC successfully captures first-mover advantage.

The I&W framework proved effective in translating qualitative risks into measurable indicators, warning thresholds, and quantitative decision-support information. By combining Monte Carlo simulation, sensitivity analysis, EMV analysis, and machine learning concepts, CIBC can strengthen monitoring, improve resource allocation decisions, and respond proactively to emerging threats and opportunities. The resulting RT&RP updates provide a more robust foundation for managing both the strategic risks and competitive opportunities associated with the acquisition.

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